

A wide-angle photograph of a desert landscape. A dirt road with visible tire tracks leads from the bottom center towards the horizon. The road is flanked by sparse desert vegetation, including small shrubs and yucca plants. In the background, large, flat-topped red rock formations (mesas) rise against a sky filled with soft, white clouds. The overall color palette is dominated by the reds of the earth and rocks, the blues of the sky, and the whites of the clouds.

Top 10 Azure Security Best Practices

Gregor Reimling

About "Gregor Reimling"



Focus

Azure Governance, Security and IaaS

Certifications

Cloud Architect & MVP for MS Azure

From

Cologne, Germany

Hobbies

Family, Community,
Worldtraveler

My Blog

<https://www.Reimling.eu>



Contact



@GregorReimling
@CloudInspires

Introduction

1. Enterprise Scale

2. Hybrid Identität

3. Identity Secure Score

4. PAW

5. Network

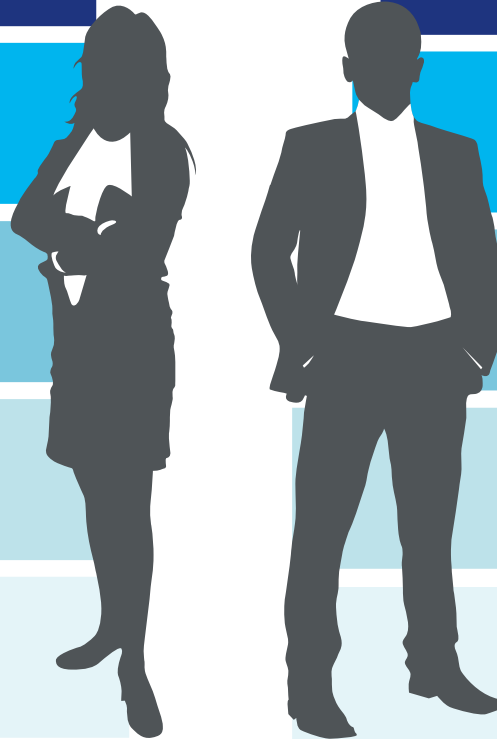
6. Defender for Cloud

7. Microsoft Sentinel

8. Azure Arc

9. Side Notes

10. Learning



Important Dates

Microsoft Azure

Search resources, services, and docs (G+)

Dashboard > Build > Cloud > Security > Security

Security Note

Effective October 1, 2022, we will begin to permanently disable Basic Authentication for Exchange Online in all Microsoft 365 tenants regardless of usage, except for SMTP Authentication. [Read more here](#)

Protect

- Conditional Access
- Identity Protection
- Security Center
- Verifiable credentials (Preview)

Manage

- Identity Secure Score
- Named locations
- Authentication methods
- Multifactor authentication
- Certificate authorities

Report

- Risky users
- Risky workload identities (preview)
- Risky sign-ins
- Risk detections

Troubleshooting + Support

- New support request

Secure Score for Identity

46.70%

Comparison

Build-Clouds	46.70%
Typical 1-1000 person company	54.10%

Score history

7 days 30 days 60 days 90 days

August 7

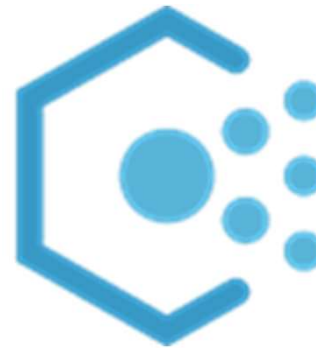
Retiring Azure AD Connect 1.x versions

Important

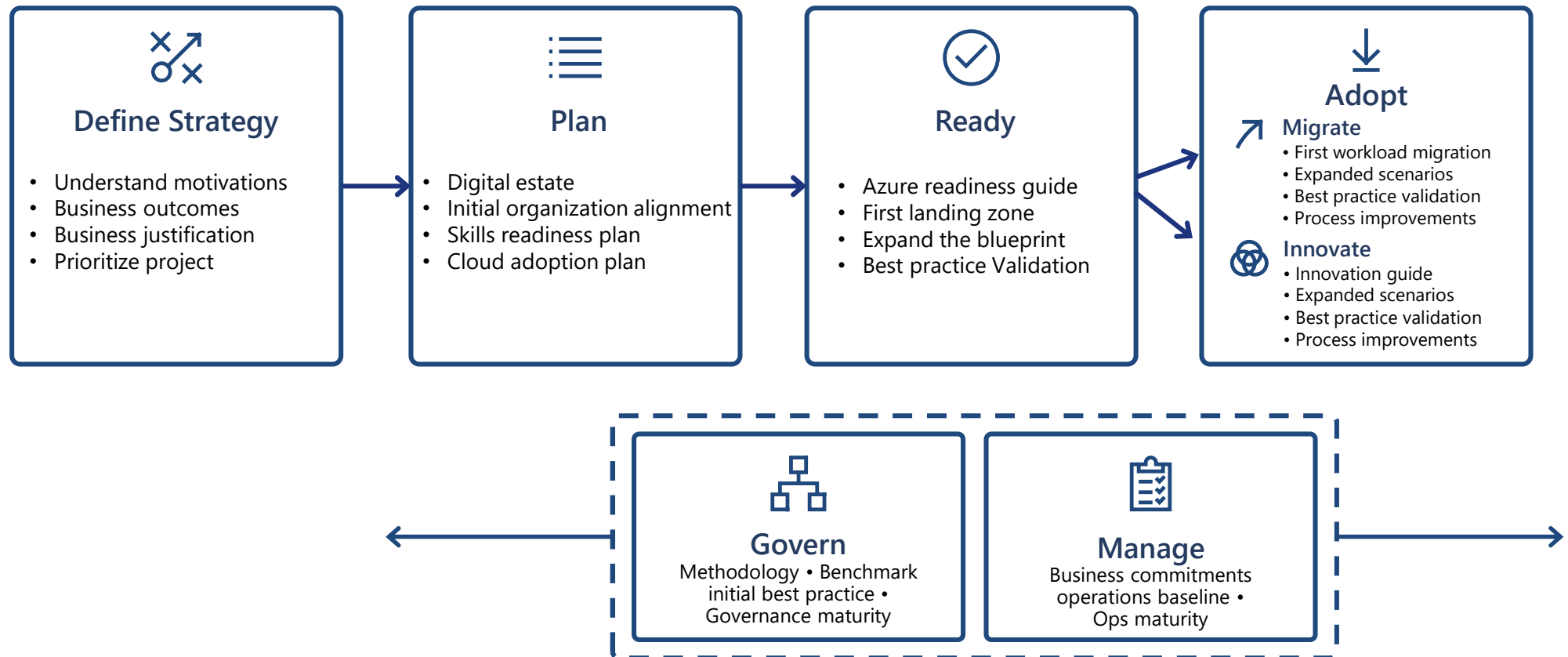
On August 31, 2022, all 1.x versions of Azure AD Connect will be retired because they include SQL Server 2012 components that will no longer be supported. Upgrade to the most recent version of Azure AD Connect (2.x version) by that date or **evaluate and switch to Azure AD cloud sync**.

Designate more than one global admin	1.64%	Low	Low
Enable Password Hash Sync if hybrid	8.20%	Low	Low
Enable policy to block legacy authentication	13.11%	Moderate	Moderate
Ensure all users can complete multi-factor au...	14.75%	High	High
Require MFA for administrative roles	16.39%	Low	Low
Enable self-service password reset	1.64%	Moderate	Moderate
Do not expire passwords	13.11%	Moderate	Low

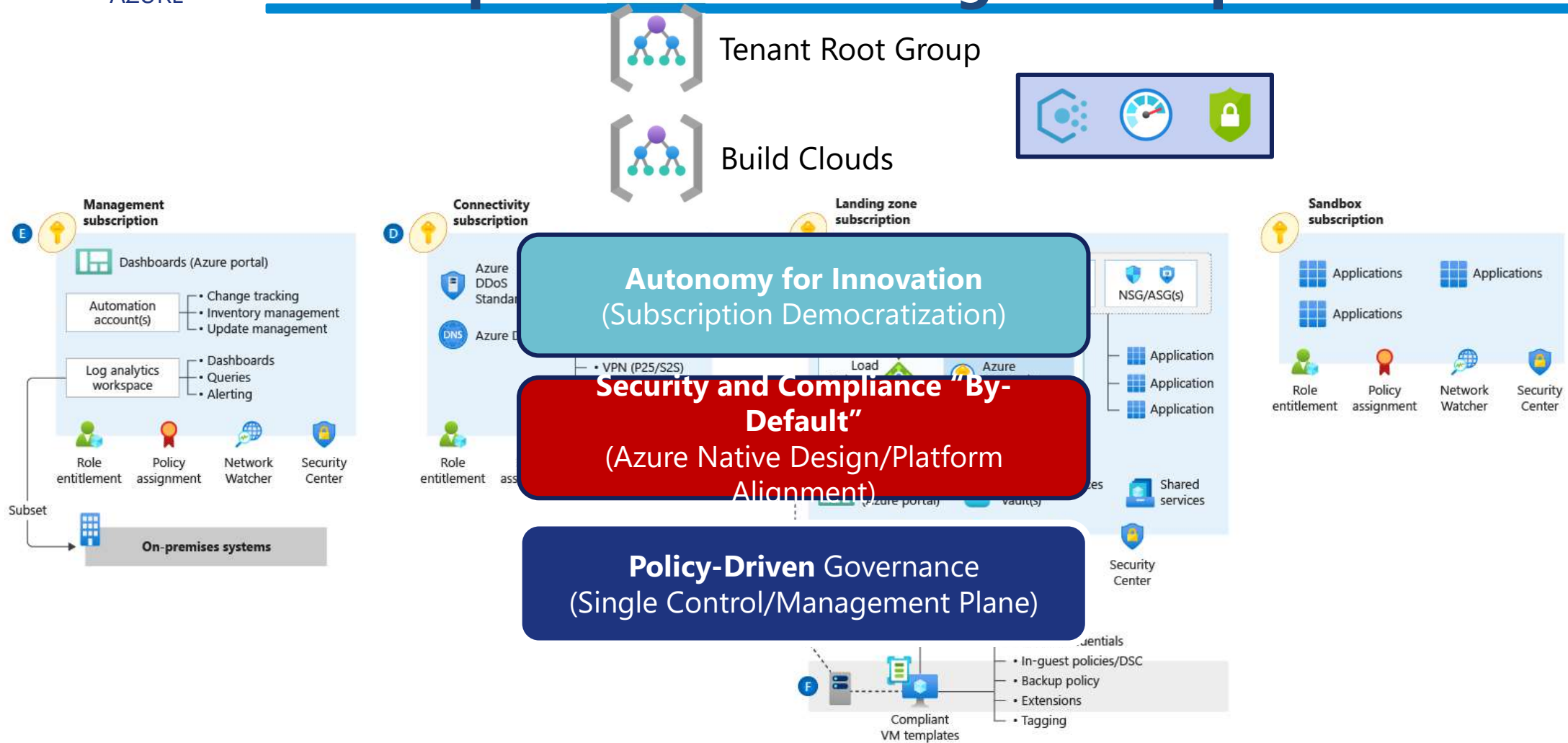
1. Enterprise Scale



MS Cloud Adoption Framework

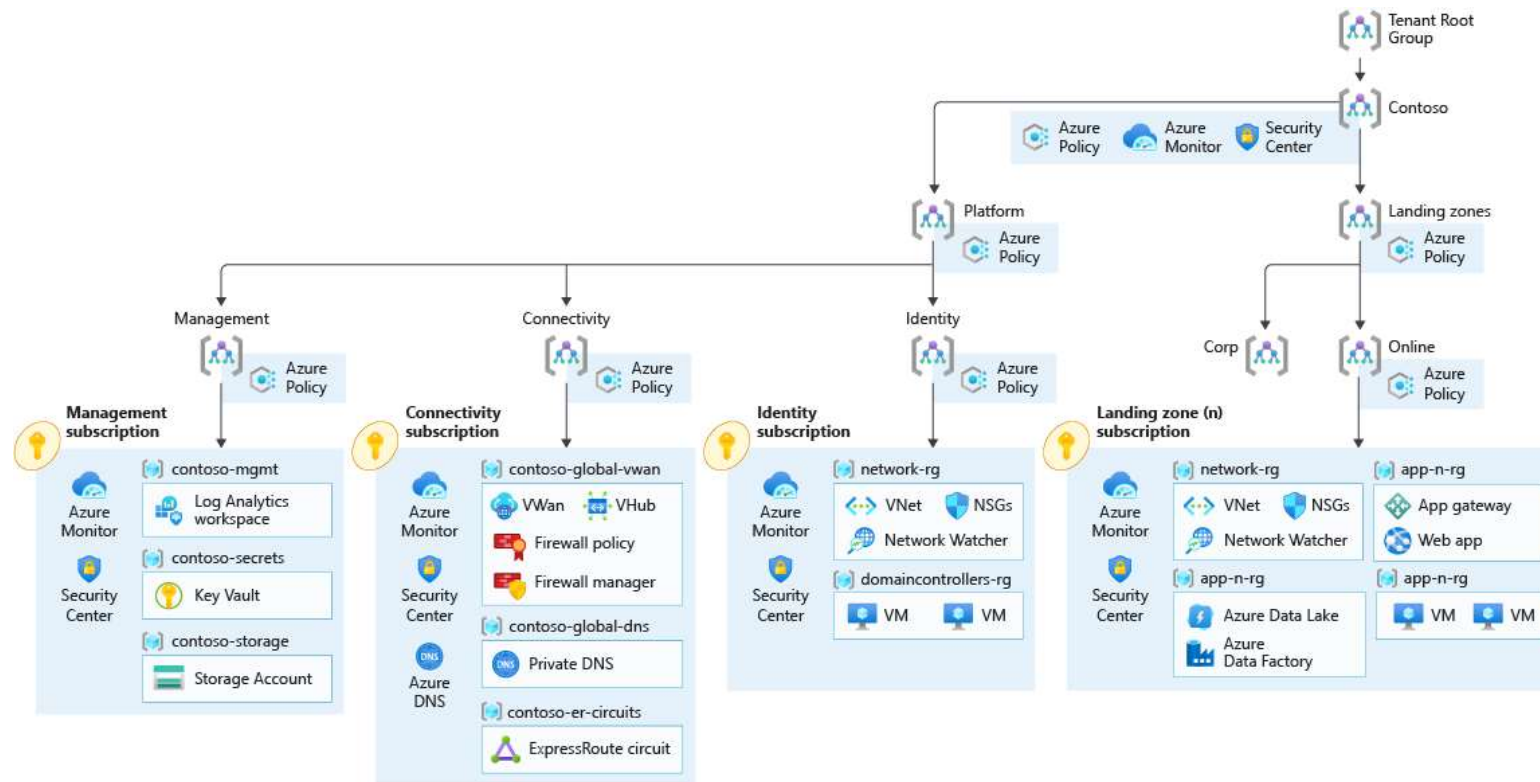


Enterprise-Scale - Design Principles



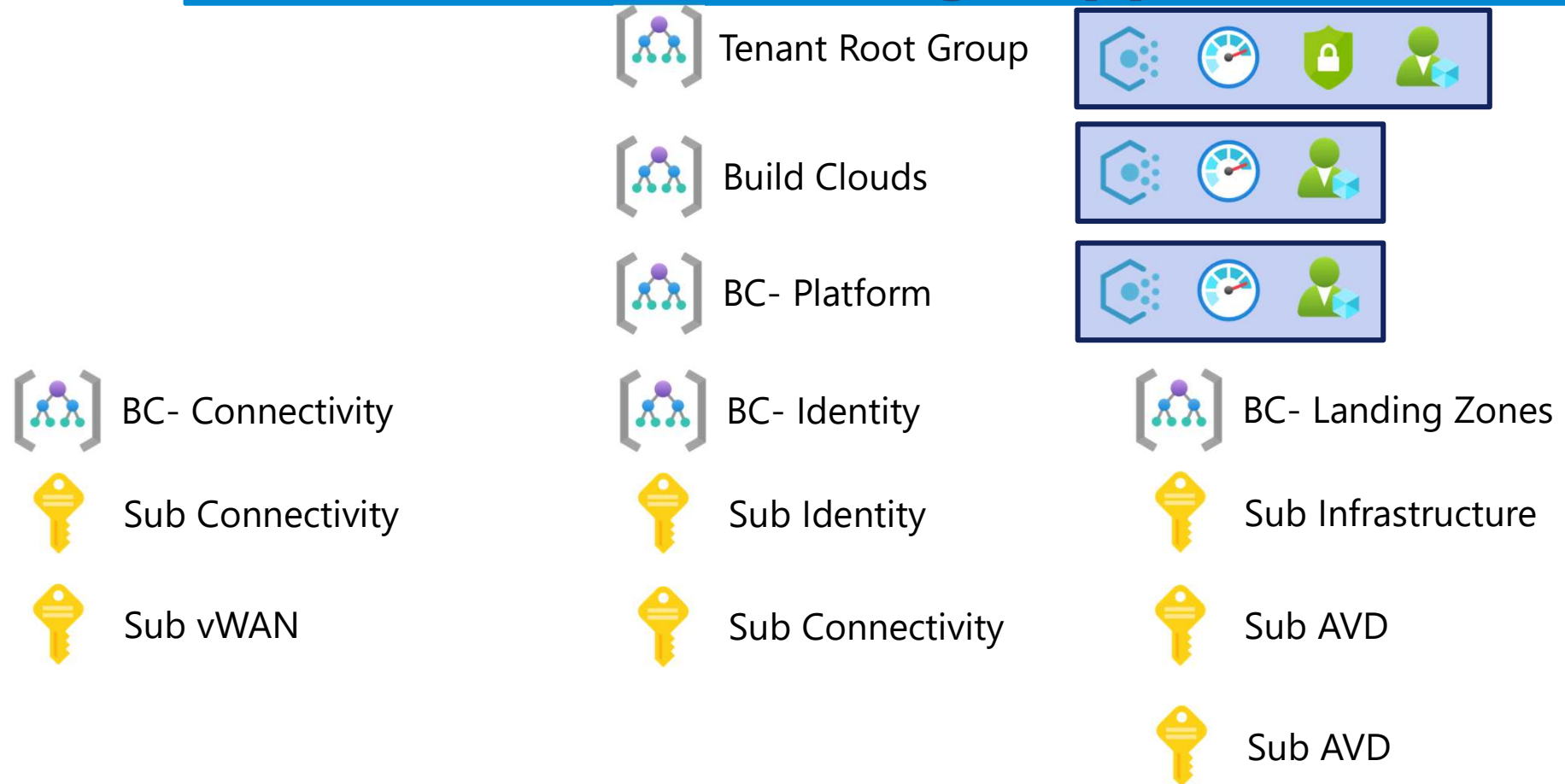
GitHub Enterprise Scale Templates

Deploy Enterprise-Scale with Azure VWAN



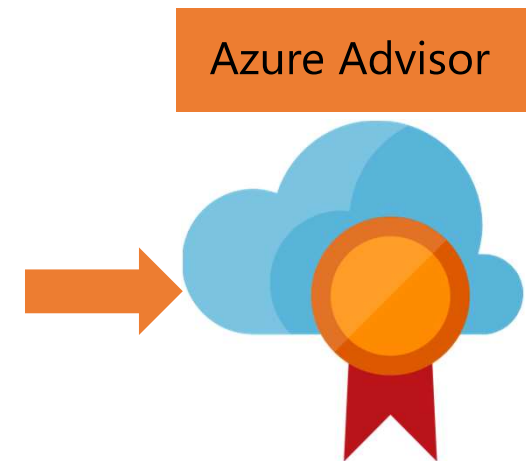
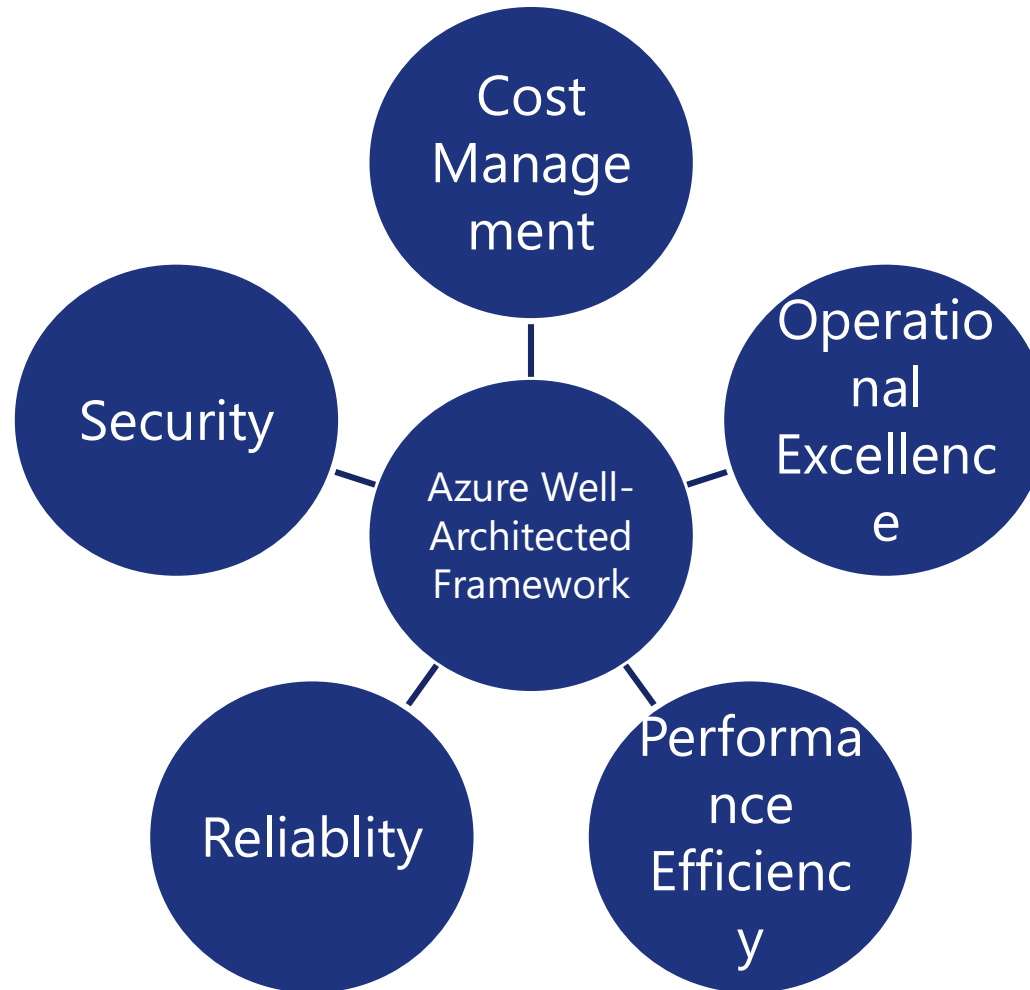
GitHub - Azure/Enterprise-Scale: The Azure Landing Zones (Enterprise-Scale) architecture provides prescriptive guidance coupled with Azure best practices, and it follows design principles across the critical design areas for organizations to define their Azure architecture

Azure Tenant Design approach



Well-architected Framework

"The Azure Well-Architected Framework is a set of guiding tenets that can be used to improve the quality of a workload."



AZURE

Demo

Dive into the Azure Portal

- Enterprise Scale
- Azure Advisor

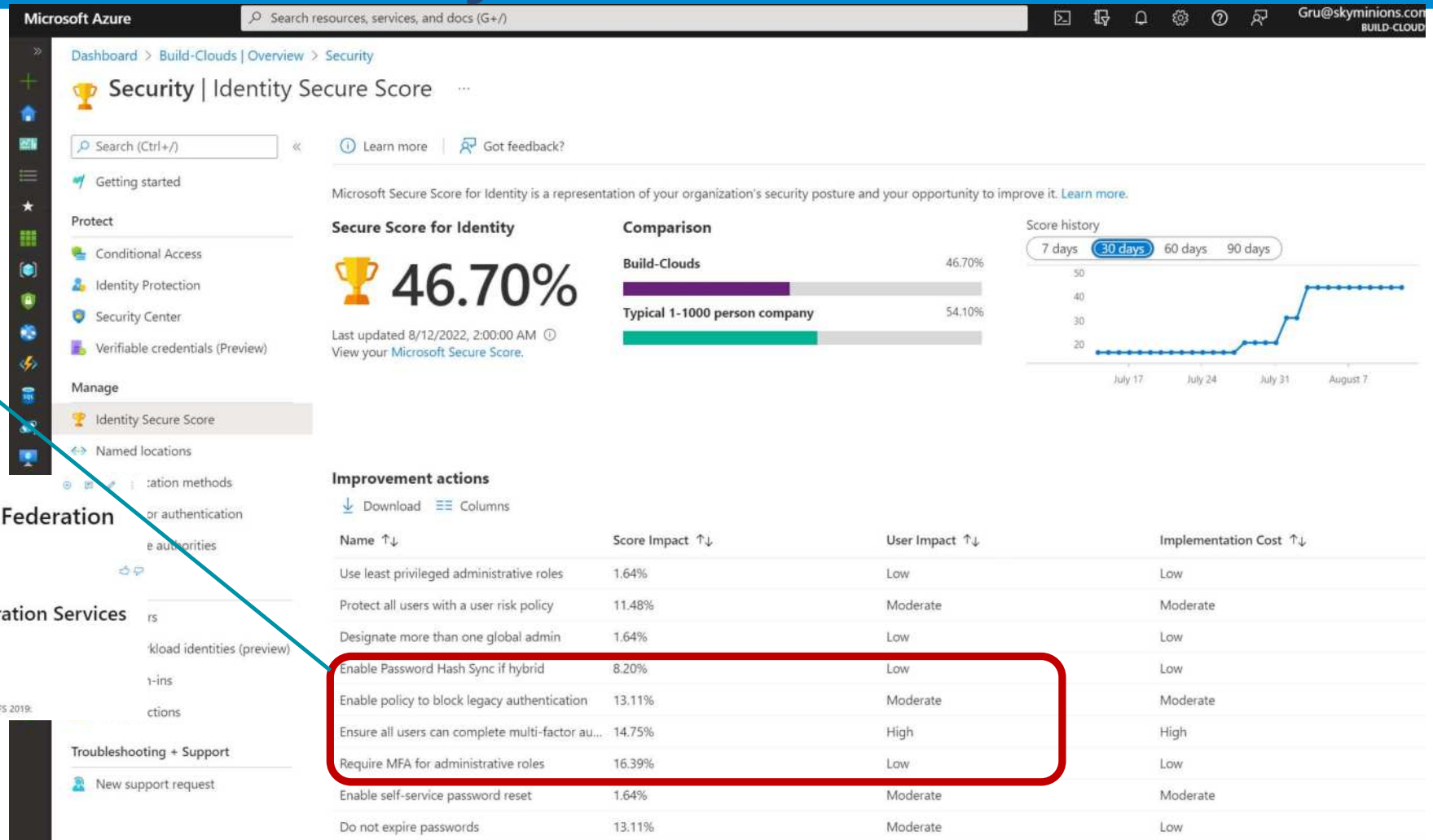


2. Hybrid Identity

AZURE

Identity Secure Score

Enable PHS
and forgot
ADFS



What's new in Active Directory Federation Services

Article • 07/05/2022 • 15 minutes to read • 20 contributors

What's new in Active Directory Federation Services for Windows Server 2019

Protected Logins

The following is a brief summary of updates to protected logins available in AD FS 2019:

Azure AD Cloud ~~Connect~~ Sync

	Connect Sync	Cloud Sync
Connect to multiple disconnected on-premises AD forests	No	Yes
Lightweight agent installation model	No	Yes
Multiple active agents for high availability	No	Yes
Connect to LDAP directories	Yes	No
Support for device objects	Yes	No
Support for device writeback	Yes	No
Support for group writeback	Yes	No
Support for Pass-Through Authentication	Yes	No

Temporary Access Pass

The screenshot displays the Microsoft Azure portal interface for configuring Temporary Access Passes. On the left, the 'Temporary Access Pass settings' page is shown with the 'Basics' tab selected. The 'ENABLE' toggle is set to 'Yes'. Under 'USE FOR:', 'Sign in' and 'Onboarding and recovery' are listed. The 'TARGET' is set to 'All users'. The 'Name' field is empty, and 'All users' is listed under 'All users'. The 'Configure' tab is also visible.

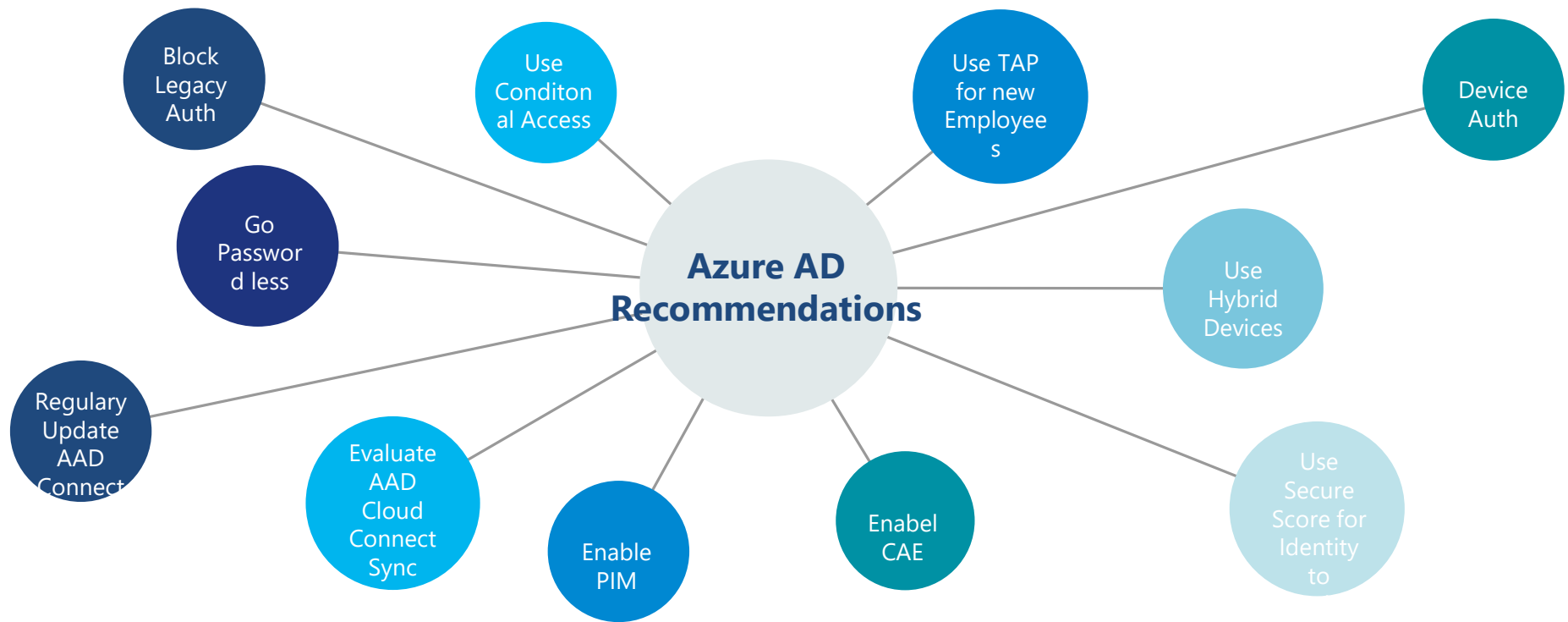
The main content area shows the 'Temporary Access Pass' settings for the user 'tap@build-clouds.com'. The 'Minimum lifetime' is set to 'Hours' (1 hour). The 'Maximum lifetime' is set to 'Hours' (4 hours). The 'Default lifetime' is set to 'Hours' (1 hour). The 'Length (characters)' is set to 8. The 'Require one-time use' toggle is set to 'No'.

Below the settings, the user is prompted to 'Enter Temporary Access Pass'. A checkbox for 'Show Temporary Access Pass' is present. A link 'Use your password instead' is provided. A 'Sign in' button is at the bottom right.

On the right, a 'Temporary Access Pass details' dialog box is open. It shows the 'Provide Pass' section with a masked pass. The 'Secure registration' section provides a link to 'https://aka.ms/mysecurityinfo'. The 'Additional information' section shows the pass is valid from 7/31/2022, 10:05:51 PM to 8/1/2022, 2:05:51 AM, and was created on 7/31/2022, 10:05:52 PM. A note at the bottom states: 'Remove lost devices from the user's account. This is especially important for devices used for user authentication.'

<https://aka.ms/mysecurityinfo>

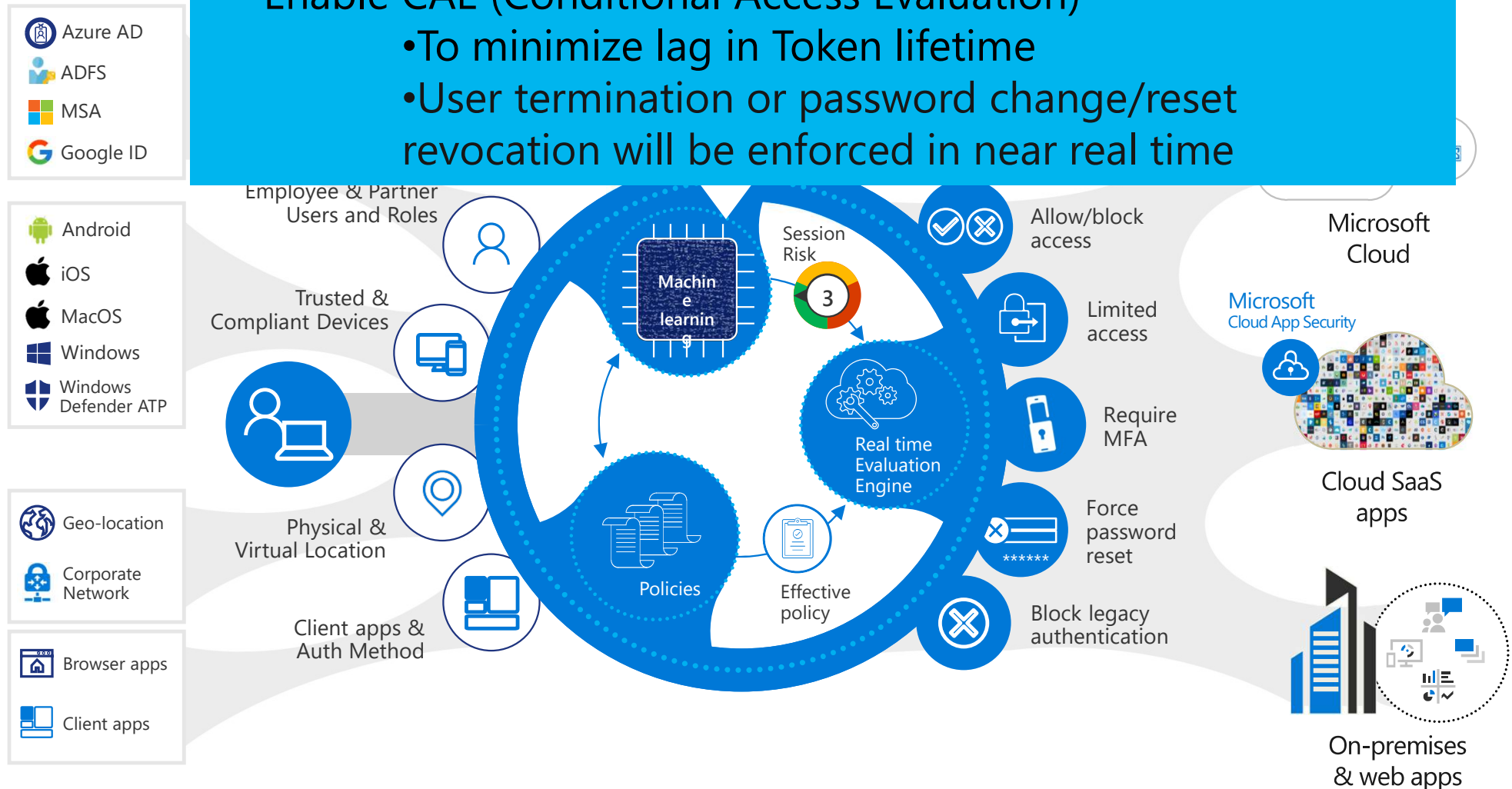
Azure AD Recommendations



3. Identity Secure Score

Identity Protection with Conditional Access

- Enable CAE (Conditional Access Evaluation)
 - To minimize lag in Token lifetime
 - User termination or password change/reset revocation will be enforced in near real time



Show following settings

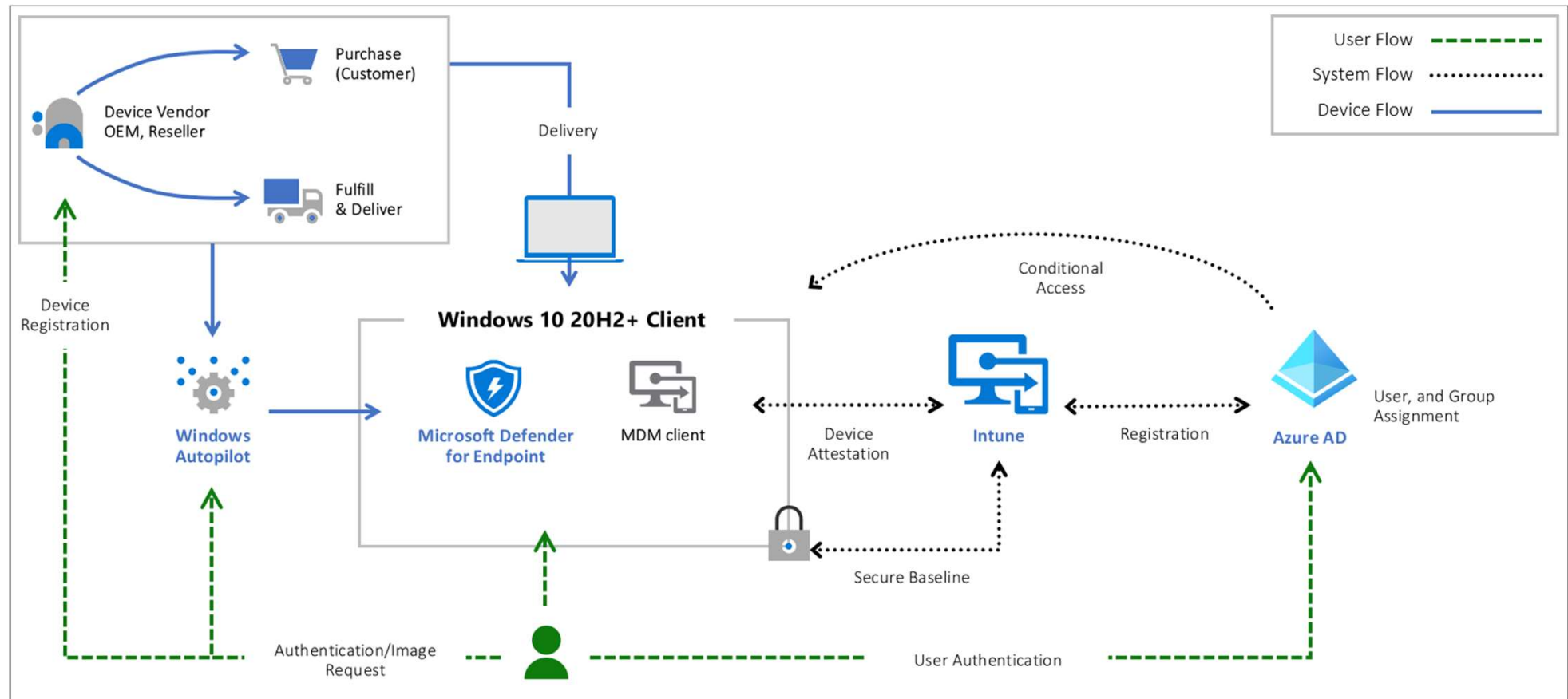
- Identity Secure Score



AZURE

4. PAW

Privileged Access Devices



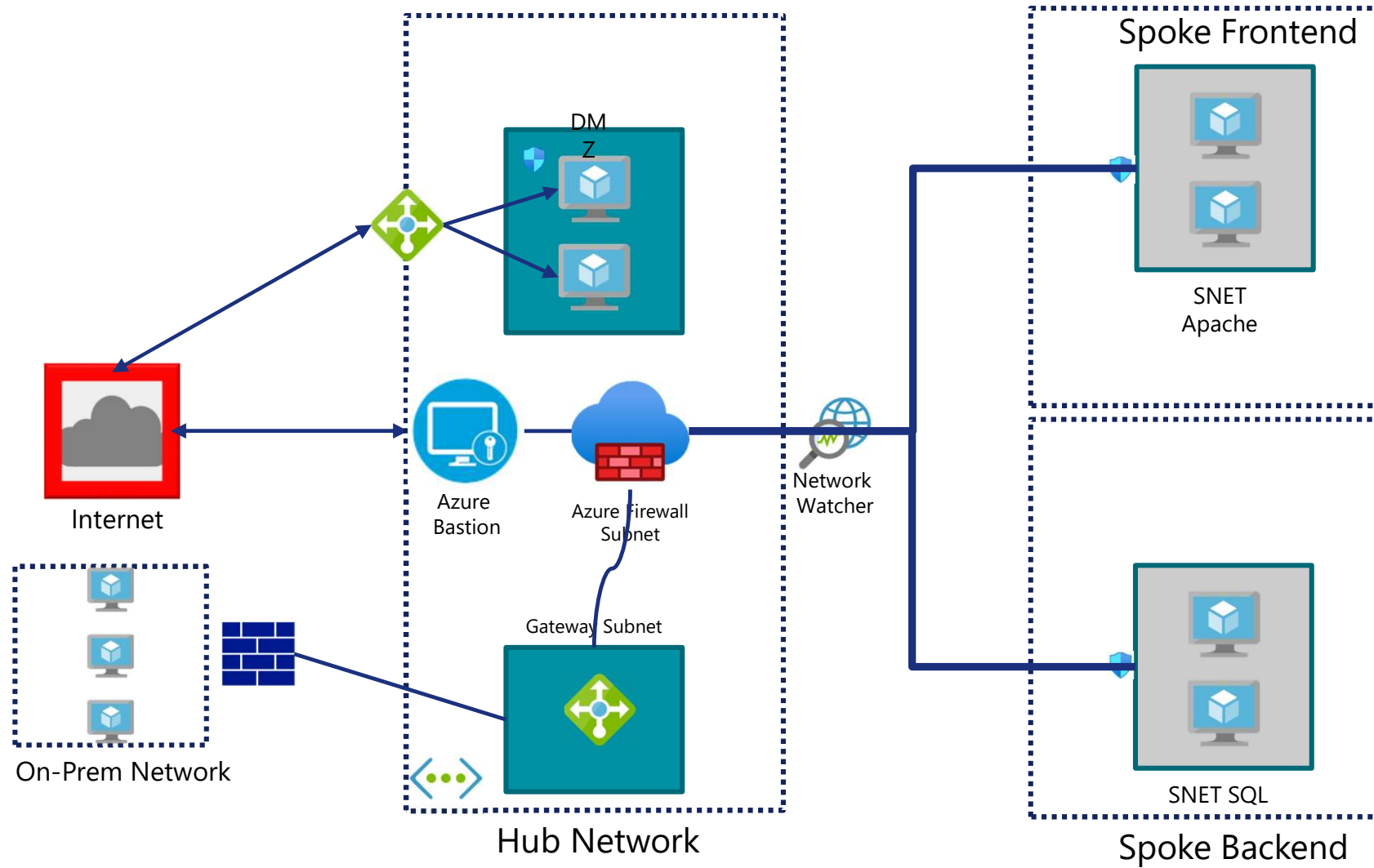
[Why are privileged access devices important | Microsoft Learn](#)

Privileged Admin Workstation

- ✓ Use dedicated VMs for manage Azure/Microsoft 365 environments
- ✓ Use this VMs only for Administration tasks
- ✓ Do not enable Internet- / Social media access on this VMs
- ✓ Enforce Device compliance via Conditional Access for this VMs
- ✓ Use AVD as PAW solution

5. Network

Network Protection



Azure IaaS Recommendations



Segmentation of Virtual Networks



Define Subnets and use NSG at Subnet Level



Use a NVA or Azure Firewall at the Hub Network



Define UDR to Route traffic over the Hub Network and Firewall



Use Azure Web Application Firewall for Internet applications



Use DDoS Protection for Web Applications

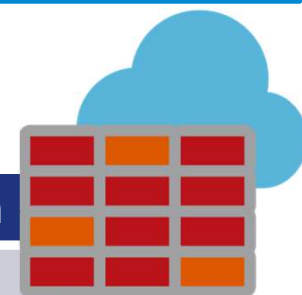


Use Azure Bastion for VM Management

AZURE

Azure Firewall Editions

New in Public Preview
(MS Ignite 2022)



Azure Firewall Basic
2 VMs fixed under the hood
Availability Zones
App FQDN Filtering Rules?
Fixed Scale
Threat Intelligence (Alert Mode only)
FQDN in Network rules
250-500MBps
Around XXX€

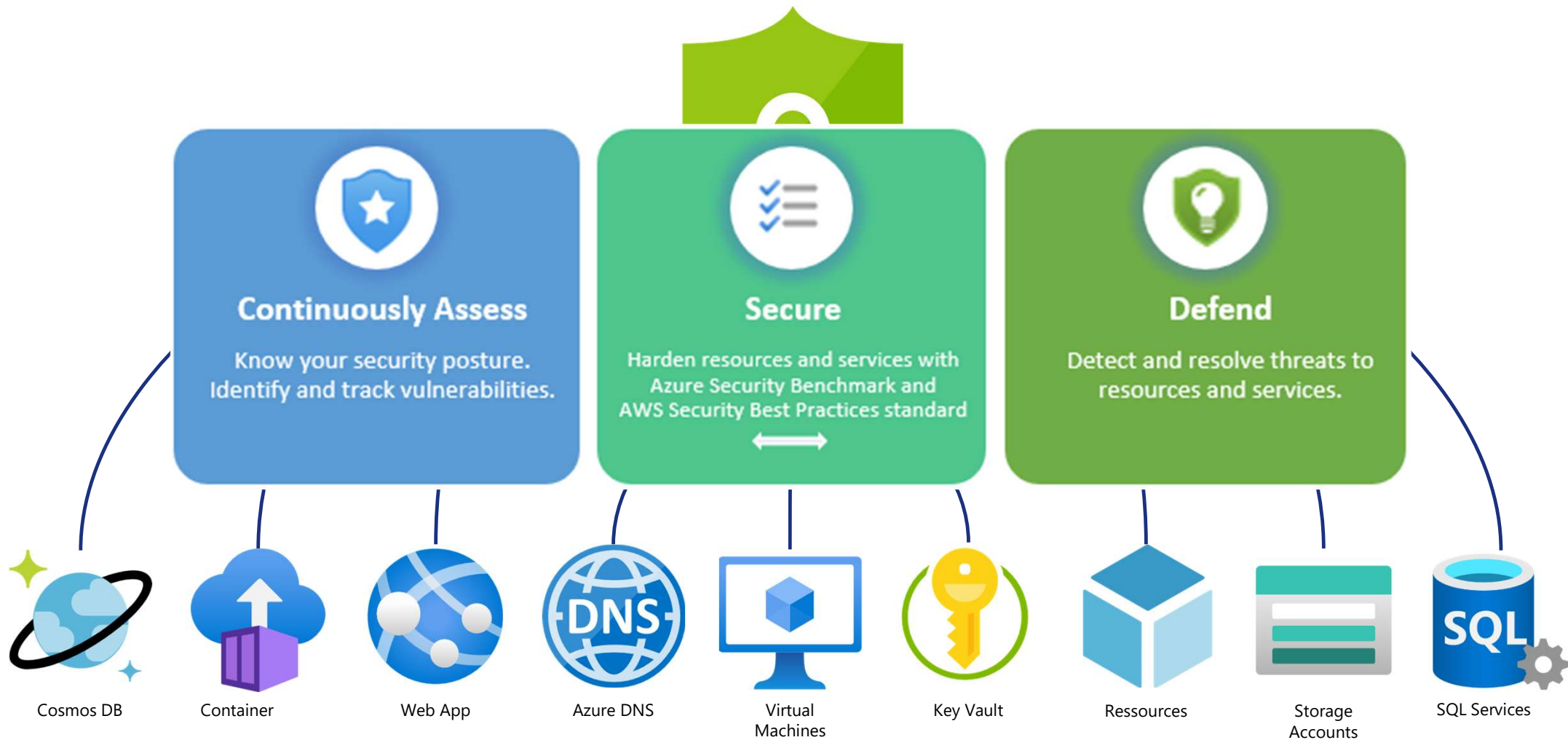
Azure Firewall Standard
Built-in high availability
Availability Zones
Application FQDN Filtering Rules
Unrestricted Cloud Scalability
Threat Intelligence
FQDN in Network rules
30GBps
901,24€ per month

Azure Firewall Premium
All from Standard +
TLS Inspection
IDPS
URL Filtering
Web categories
FQDN in Network rules
30GBps
1.262,29€ per month



6. MS Defender for Cloud

Microsoft Defender for Cloud



AZURE

MS Defender for Cloud



Security posture
& compliance

Secure score

Asset management

Policy



Server protection
(Microsoft Defender for Cloud for VMs)

Threat detection

VA (power by Qualys)

Application control



Automation &
management at scale

Automation

SIEM integration

Export

Defender for Servers Plans

	Plan 1	Plan 2
Unified View	Yes	Yes
Automatic MDE provisioning	Yes	Yes
MS Threat and Vulnerability management	Yes	Yes
Security Policy and Regulatory Compliance	No	Yes
Integrated Vulnerability by Qualys	No	Yes
Log Analytics 500MB free data ingestion per day	No	Yes
Threat detection	No	Yes
Adaptive application control	No	Yes
File integrity monitoring	No	Yes
Just-in-Time VM access	No	Yes
Adaptive Network hardening	No	Yes
Docker host hardening	No	Yes
Fileless attack detection	No	Yes
Price	5\$ per Server	15\$ per Server

Dive into the Azure Portal

- Microsoft Defender for Cloud



AZURE

Dashboard > Update management center (Preview) | Machines

Update management center (Preview) | Machines

Search (Ctrl+F)

Overview
Getting started

Manage
Machines
History
Support > troubleshooting
New Support Request

Filter by name
Subscription: M00000000000000000000000000000000
Resource group: AB
Resource type: AB
Location: AB
OS: AB
Patch orchestration: AB
Periodic assessment: AB
Status: AB
Tags: AB

Enable periodic assessment using Azure Policy or Update settings to regularly check for updates. Enable now >

Total machines: 9
No updates data: 9
No updates available: 0
Updates available: 0
Reboot Required: 0

Showing 8 of 9 records
Select all

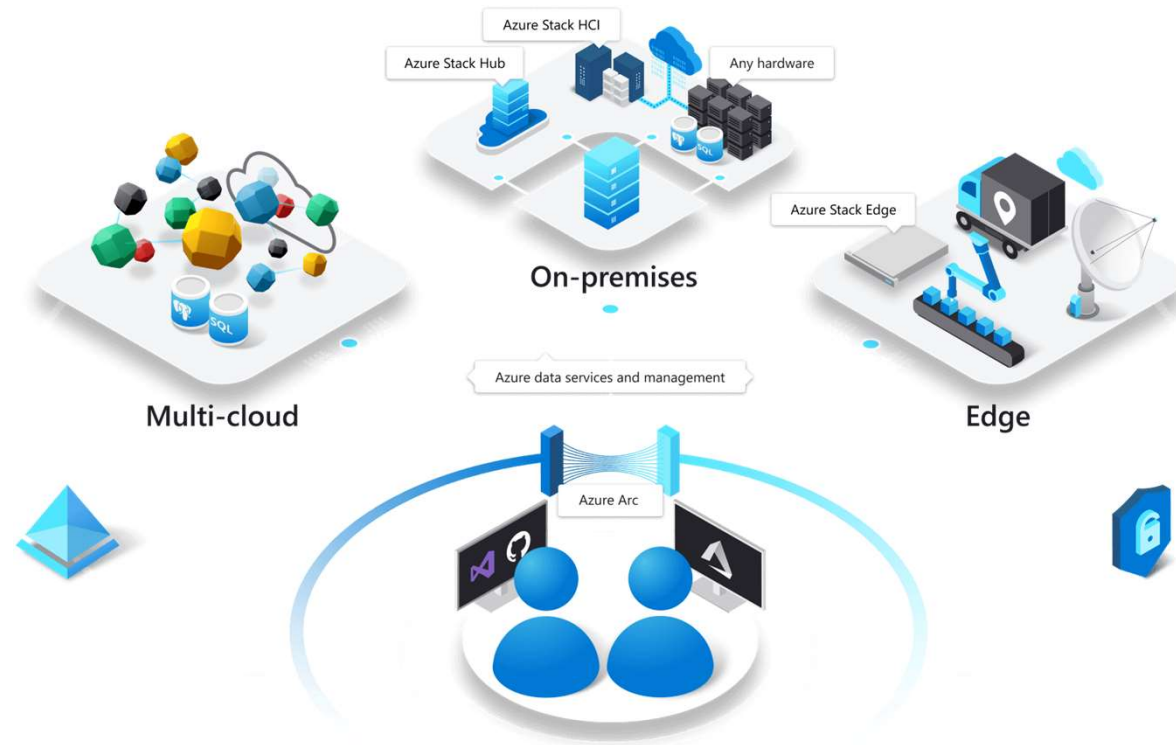
Name	Update status	Operating system	Resource type	Patch orchestration	Periodic assess...	Associated sche...	Status
MSDN00000000000000000000000000000000	No updates data	Windows	Azure Virtual Machine	Automatic by OS	No	-	VM dedicated
AVD PROD-01	No updates data	Windows	Azure Virtual Machine	Automatic by OS	No	-	VM dedicated
MSD-01	No updates data	Windows	Azure Virtual Machine	Unknown	No	-	VM dedicated
AZVMAG00000000000000000000000000000000	No updates data	Windows	Azure Virtual Machine	Automatic by OS	No	-	VM dedicated
MSLump-001	No updates data	Windows	Arc-enabled server	N/A	No	-	Expired
MSDHub01	No updates data	Windows	Arc-enabled server	N/A	No	-	Connected
MSD001	No updates data	Windows	Arc-enabled server	N/A	No	-	Connected
AZVMAG00000000000000000000000000000000	No updates data	Windows	Azure Virtual Machine	Automatic by OS	No	-	VM running
AZVMACK	No updates data	Linux	Azure Virtual Machine	Image Default	No	-	VM running



7. Microsoft Sentinel

8. Azure Arc

Azure Arc



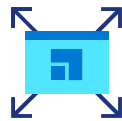
AZURE

Azure Arc: *at a high level*

Bring Azure services and management to any infrastructure, anywhere



Run Azure data
services anywhere



Extend Azure management
across your environments



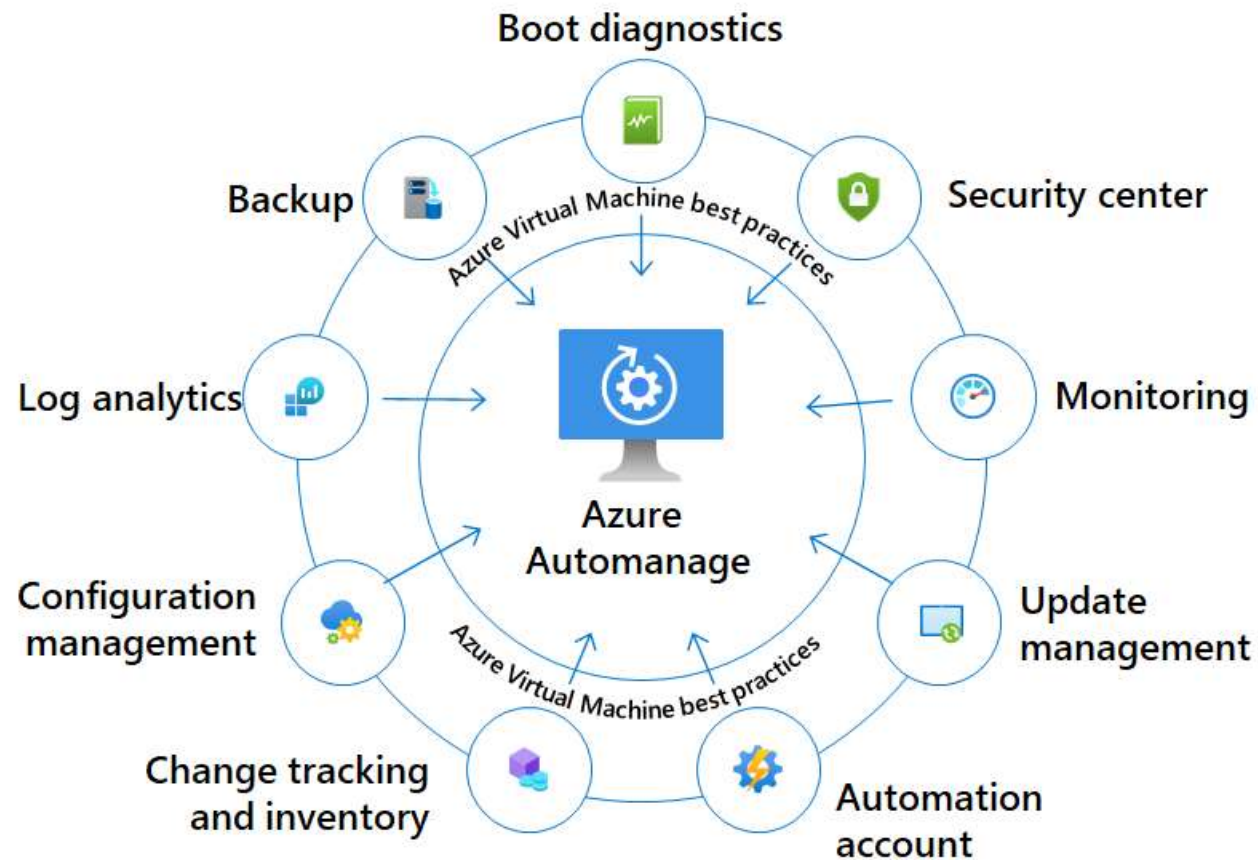
Adopt cloud
practices on-premises



Implement Azure
security anywhere

Azure Arc is a set of technologies that extends Azure management and enables Azure services to run across on-premises, multi-cloud, and edge

Azure Automanage



AZURE

Update Management Center (preview)

New solution for centrally Update Management accross different environments

No dependencys to Log Analytics Agent

Fully support for Azure Arc managed VMs

Support Windows and Linux Vms

Support automatic VM guest patching

Support Hot patching

Is in preview wait for production until release going to GA



Cloud Security Explorer



Microsoft Defender for Cloud | Cloud Security Explorer (Preview) ...

Showing 6 subscriptions

[Guides & Feedback](#)[Share query link](#)

General

Overview

Getting started

Recommendations

Security alerts

Inventory

Cloud Security Explorer (Preview)

Workbooks

Community

Diagnose and solve problems

Cloud Security

Security posture

Regulatory compliance

Workload protections

Firewall Manager

DevOps Security (Preview)

Defender CSPM plan was enabled recently on at least one of your subscriptions/accounts. It could take up to one day until all data will be available



What would you like to search?

Virtual machines

+

Clear all

AND

Has

Vulnerabilities

+

Where

Severity

Equals

High

Remove

Has

Insight

+

Where

Title

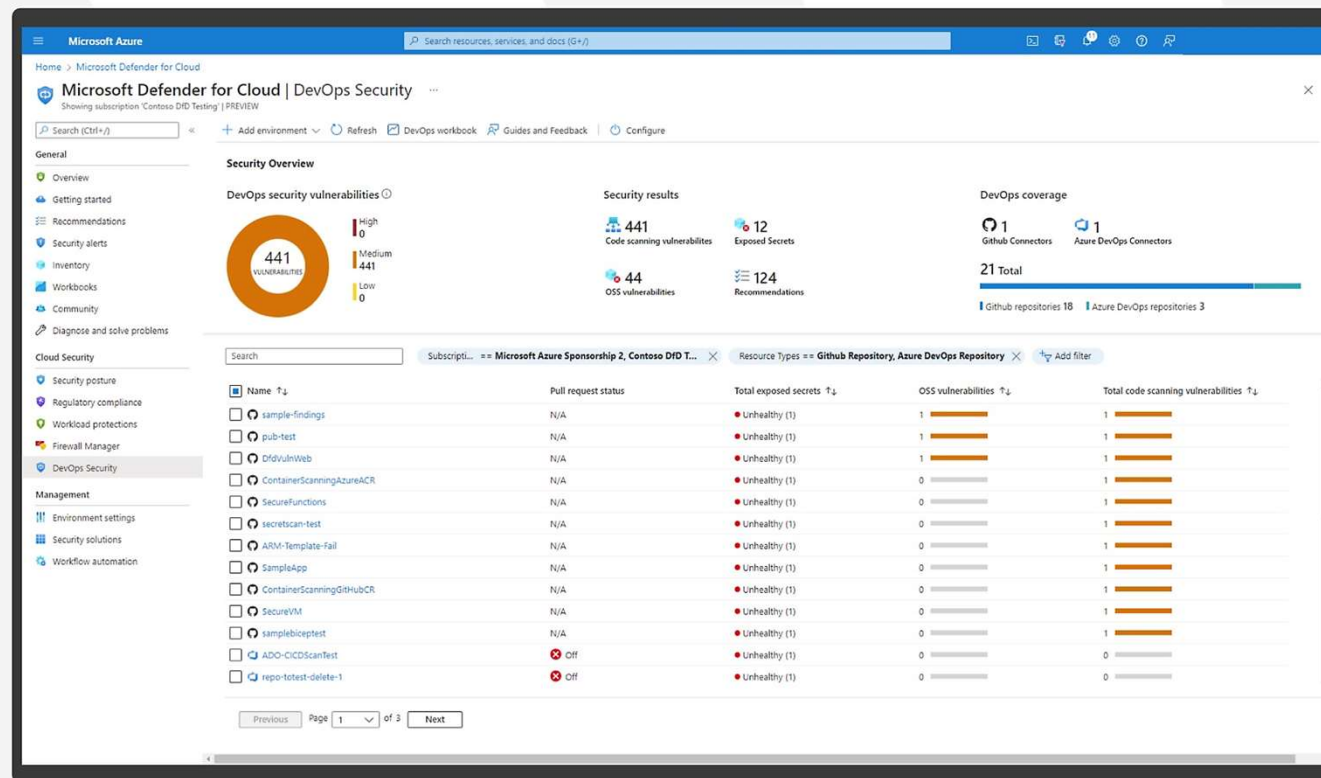
Equals

exposed to the...

Remove

Scope : MVP BC LZ Infrastructure (c7ce5f8f-0180-4dd2-b44b-2)

Defender for DevOps



[Join Our Security Community - Microsoft Community Hub](#)

Learning

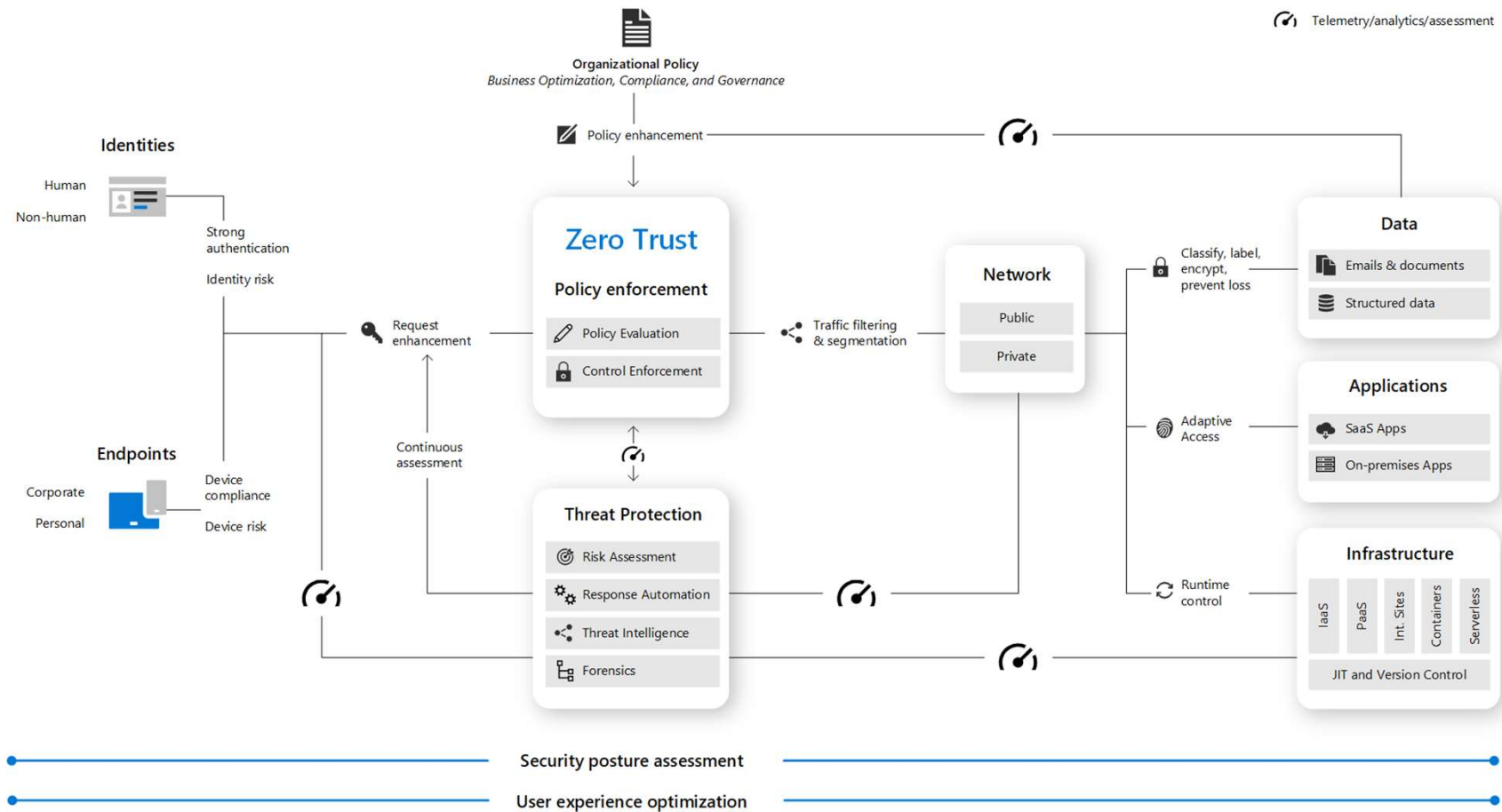
Popular learning paths and modules

LEARNING PATH Microsoft Azure Fundamentals: Describe cloud concepts 🕒 52 min Azure • Administrator • Beginner 🏆 ⊕ Save	LEARNING PATH Microsoft Azure Data Fundamentals: Explore core data concepts 🕒 45 min Azure • Data Analyst • Beginner 📊 ⊕ Save	MODULE Discuss Azure fundamental concepts 🕒 24 min ⭐⭐⭐⭐⭐ 4.8 (123,249) Azure • Administrator • Beginner 🗂️ ⊕ Save
MODULE Explore fundamentals of data visualization 🕒 38 min ⭐⭐⭐⭐⭐ 4.7 (4,086) Azure • Administrator • Beginner 📊 ⊕ Save	MODULE Introduction to Azure fundamentals 🕒 43 min ⭐⭐⭐⭐⭐ 4.8 (202,694) Azure • Administrator • Beginner 🗂️ ⊕ Save	MODULE Describe core Azure architectural components 🕒 27 min ⭐⭐⭐⭐⭐ 4.8 (88,090) Azure • Administrator • Beginner 🗂️ ⊕ Save
LEARNING PATH Microsoft Azure Data Fundamentals: Explore relational data in Azure 🕒 1 hr 13 min Azure • Data Analyst • Beginner 📊 ⊕ Save	MODULE Introduction to Microsoft Power Platform 🕒 36 min ⭐⭐⭐⭐⭐ 4.7 (37,807) Microsoft Power Platform • Business Analyst • Beginner 🗂️ ⊕ Save	LEARNING PATH Microsoft Azure Data Fundamentals: Explore non-relational data in Azure 🕒 1 hr 9 min Azure • Data Analyst • Beginner 📊 ⊕ Save

[Training | Microsoft Learn](#)

[Join Our Security Community - Microsoft Tech Community](#)

Zero Trust architecture



Microsoft Cybersecurity Reference Architectures (MCRA)

Capabilities

What cybersecurity capabilities does Microsoft have?



Azure Native Controls

What native security is available?



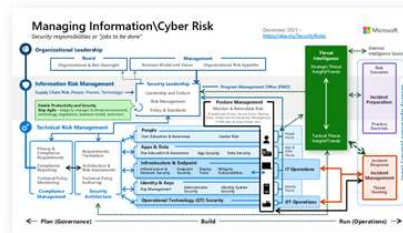
Attack Chain Coverage

How does this map to insider and external attacks?



People

How are roles & responsibilities evolving with cloud and zero trust?



Zero Trust User Access

How to validate trust of user/devices for all resources?



Security Operations

How to enable rapid incident response?



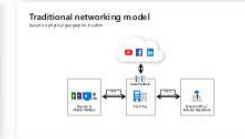
Multi-Cloud & Cross-Platform

What clouds & platforms does Microsoft protect?



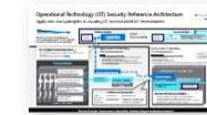
Secure Access Service Edge (SASE)

What is it? How does it compare to Zero Trust?



Operational Technology

How to enable Zero Trust Security for OT?



Top 5 Must have Settings

1. Enable Multifactor Authentication
 2. Enable and Integrate Conditional Access
 3. Use Azure Advisor recommendations
 4. Cloud Security Posture Management is needed
-

Conclusion

1. Enterprise Scale

2. Identity Secure Score

3. Use WHfB and TAP

4. Azure AD Cloud Sync

5. PAW

6. Network

7. Defender for Cloud

8. Microsoft Sentinel

9. Side Notes

10. Learning



- [Reimling.eu – Microsoft will disable Basic auth – What this means and what you have to do](#)
- [Block legacy authentication - Azure Active Directory - Microsoft Entra | Microsoft Docs](#)
- [Deprecation of Basic authentication in Exchange Online | Microsoft Docs](#)
- [Common Conditional Access policies - Azure Active Directory - Microsoft Entra | Microsoft Learn](#)
- [Configure a TAP in Azure AD to register Passwordless authentication - Microsoft Entra | Microsoft Docs](#)
- [Azure AD Connect: Version release history - Microsoft Entra | Microsoft Docs](#)
- [Zero Trust security in Azure | Microsoft Docs](#)
- [Enterprise-Scale/README.md at main · Azure/Enterprise-Scale · GitHub](#)
- [Azure Active Directory passwordless sign-in - Microsoft Entra | Microsoft Docs](#)
- [Azure Arc | Microsoft Learn](#)
- [Update management center \(preview\) overview | Microsoft Docs](#)
- [Microsoft Cybersecurity Reference Architectures - Security documentation | Microsoft Learn](#)
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- [Managed identities for Azure resources - Microsoft Entra | Microsoft Learn](#)
- [Microsoft Certified: Azure Security Engineer Associate - Certifications | Microsoft Learn](#)

About "Gregor Reimling"



Thank You



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Blog

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